INCH-POUND
MIL-H-21623B(SH)
30 April 1990
SUPERSEDING
MIL-H-21623A(SH)
5 February 1965
(See 6 5)

MILITARY SPECIFICATION

HOLDERS, RADIAC SAMPLE

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers small metal planchets and cups to be used in radiological monitoring work.

1 2 Classification The holders shall be of the following types as specified (see 6.2).

Type I - Holder, radiac sample, aluminum.
Type II - Holder, radiac sample, (planchet)

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2 1 1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL
PPP-B-636 - Boxes, Shipping, Fiberboard.
PPP-T-97 - Tape, Packaging/Industrial, Filament Reinforced

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to. Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MILITARY
MIL-P-116 - Preservation, Methods of

STANDARDS

MILITARY
MIL-STD-129 - Marking for Shipment and Storage

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, BLDG 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094)

2 2 Non-Government publications The following document(s) form a part of this document to the extent specified herein Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)
A 240 - Standard Specification for Heat-Resisting Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)
AMS 5503 - Steel Sheet, Strip, and Plate, Corrosion Resistant 17Cr (SAE 51430) Annealed.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

UNIFORM CLASSIFICATION COMMITTEE AGENT
Uniform Freight Classification Ratings, Rules and Regulations

(Application for copies should be addressed to Uniform Classification Committee Agent, Tariff Publication Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

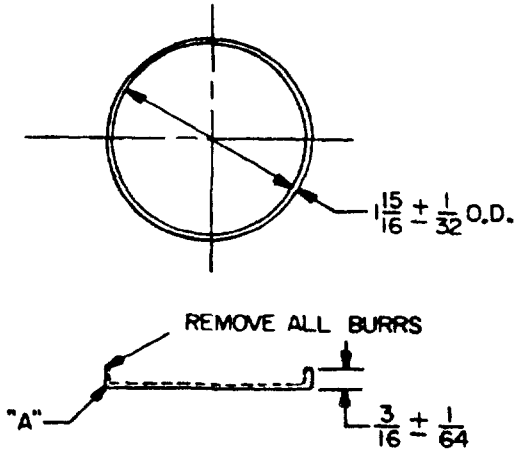
2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3 REQUIREMENTS

3 1 Material and size

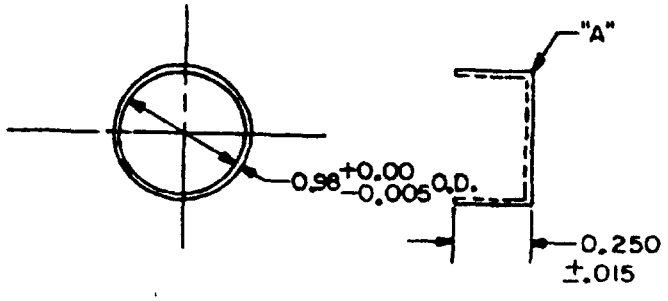
(a) Type I holder shall be commercial grade aluminum alloy with size as shown on figure 1



Metal thickness - 0.015
All dimensions in inches
"A" tooling radius to be about 1/16

FIGURE 1 Type I aluminum radiac sample holder.

(b) Type II holder shall be commercial grade, type 430, corrosion-resisting steel with size as shown on figure 2 The sheet material used to tabulate the holders shall conform to either ASTM A 240 or SAE AMS 5503 and shall have received a No. 2B finish (as defined in ASTM A 240) on both sides.



Finish - Polish
Metal thickness - 0.015
All dimensions in inches
"A" tooling radius to be about 1/32

FIGURE 2. Type II corrosion resisting steel radiac sample holder.

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3 2 Radioactivity Residual radioactivity of the holders shall not be detectable above normal background when tested as specified in 4 3

3 3 Recovered materials Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use The term "recovered materials" means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specifically specified

3 4 Workmanship. The holders shall be uniform in quality and condition; clean, sound, smooth, and free from buckles, blisters and other injurious defects within the limits consistent with best commercial practice Discoloration due to heat treatment shall not be cause for rejection

4 QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract (see 6.3). Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material

4.2 Sampling Sampling for radioactivity inspection, visual and dimensional examinations, and packaging inspections shall be in accordance with table I. The lot size is defined as the quantity of items offered for delivery in partial fulfillment of a Government contract or complete fulfillment if there is only one delivery.

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TABLE I. Sampling for visual and dimensional examinations

Lot size	Sample size	Accept	Reject 1/ 2/ 3/
3 - 15	3	0	1
16 - 25	5	0	1
26 - 50	8	0	1
51 - 90	13	1	2
91 - 150	20	2	3
151 - 280	32	3	4
281 - 500	50	8	9
501 - 1200	80	14	15
1201 - 3200	125	21	22
3201 - Over	200	21	22

1/ All defective items must be replaced with acceptable items prior to lot acceptance.
2/ Inspect sample size until reject criteria is reached
3/ Rejected lots may be screened and resubmitted for inspection and retest.

4.3 Radioactivity inspection. The residual radioactivity level of the holders shall be measured to determine that the activity level is not detectable above normal background radiation as follows. Each selected sample shall contain less than 1.0 picocurie of equivalent strontium - 90 radioactivity (at the 95 percent confidence level) when individually counted using a low background beta - proportional counter which has been calibrated with a strontium - 90 standard

4.4 Visual and dimensional examinations. Visual and dimensional examinations shall be made in accordance with table I to determine the quality of dimensions and finish as specified on figures 1 and 2, and in 3 4

4.5 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition.)

5 1 Domestic shipment.

5.1.1 Preservation. Preservation shall be sufficient to afford adequate protection against corrosion, deterioration and physical damage during shipment from the supply source to the using activity and until early use and may conform to the supplier's commercial practice when such meets these requirements.

5.1.2 Packing. Packing shall be accomplished in a manner which will insure acceptance by common carrier at the lowest rate and will afford protection against physical or mechanical damage during direct shipment from the supply source to the using activity. The shipping containers or method of packing shall conform to the

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Uniform Freight Classification Ratings, Rules and Regulations, or other carrier regulations as applicable to the mode of transportation and may conform to the supplier's commercial practice when such meets these requirements

5 1.3 Marking Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the contractor's commercial practice The information shall include nomenclature, national stock number or manufacturer's part number, contract or order number, contractor's name and destination.

5 2 Domestic shipment and storage or overseas shipment The requirements, and levels of preservation, packing, and marking for shipment shall be as specified (see 6.2).

5 2 1 Level A preservation Level A preservation shall be in accordance with method III of MIL-P-116

5 2.2 Level C, types I and II The preservation of aluminum holders shall be in accordance with the manufacturers commercial practice

5.2.3 Packing, level A and B, types I and II Ten unit packages of holders shall be packed in a snug fitting box conforming to style RSC, type SF, weather resistant grade V3s of PPP-B-636. All joints and corners of the box shall be sealed with a minimum of 1-1/2 inch wide tape conforming to PPP-T-97. Strapping shall be with tape conforming to type III of PPP-T-97 and the appendix thereto.

5.2.4 Level C, types I and II. The holders, preserved as specified (see 5.2.2), shall be packed in containers which will insure acceptance by common carrier and safe delivery at destination Shipping containers shall comply with the Uniform Freight Classification Ratings, Rules and Regulations or other regulations as applicable to the mode of transportation.

5 2 5 Marking In addition to any special marking (see 6.2), interior and exterior shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Holders will be used for measurement of radioactive samples on nuclear powered ships.

6 2 Acquisition requirements. Acquisition documents must specify the following:

(a) Title, number, and date of this specification.
(b) Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 1 and 2.2).
(c) Type required (see 1.2).
(d) Level of preservation, packing, and marking required if other than specified (see 5 2 and 5.2.5).

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6 3 Consideration of data The following Data Item Descriptions (DID's) should be considered when this specification is applied on a contract The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that DID's are tailored to reflect the requirements of the specific acquisition To ensure correct contractual application requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27 475-1 exempts the requirement for a DD Form 1423

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
4 1 1	DI-E-2121	Certificate of compliance	----

The above DID's were those cleared as of the date of this specification The current issue of DOD 5010 12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423

6 4 Subject term (key word) listing.

Aluminum
Corrosion resisting steel
Planchet
Residual radioactivity

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Preparing activity:
Navy - SH
(Project 6665-N594)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE. This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-H-21623B(SH)	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE
HOLDERS, RADIAC SAMPLE

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact (TPOC): Mr. D. Schaeffer (CEL-R1) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON TPOC: 703-602-7383
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office

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